Alex Rivers

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Professional Summary

Innovative **AI Engineer** with 5+ years of experience designing and deploying scalable machine learning solutions. Expert in bridging the gap between research-level models and production-ready applications. Specializing in **Computer Vision** and **Natural Language Processing**, with a proven track record of optimizing model performance and reducing latency in containerized environments.

Technical Skills

- **Languages:** Python (Expert), C++, SQL, Bash
- **AI Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn
- **Computer Vision:** OpenCV, MediaPipe, YOLOv8, Image Segmentation
- **Deployment & DevOps:** Docker, Kubernetes, AWS (SageMaker/EC2), CI/CD Pipelines
- **Data Engineering:** Pandas, NumPy, Spark, Kafka

Professional Experience

Senior AI Engineer | NeuralPath Tech *January 2021 – Present*

- Architected a real-time object detection system using **OpenCV** and **PyTorch**, improving processing speed by **35%** for edge devices.
- Containerized the entire machine learning pipeline using **Docker**, ensuring seamless deployment across heterogeneous cloud environments.

- Led a team of 4 to develop a generative AI chatbot that reduced customer support tickets by **20%** within the first quarter.
- Optimized deep learning models for production, reducing memory footprint by **150MB** without sacrificing accuracy.

Machine Learning Engineer | DataStream Solutions *June 2018 – December 2020*

- Developed predictive maintenance models using **TensorFlow**, saving enterprise clients an estimated **$2M** in annual hardware costs.
- Engineered automated data labeling scripts in **Python**, reducing manual annotation time by **60%**.
- Implemented robust ETL pipelines to handle petabyte-scale datasets for model training.

Key Projects

VisionGuard | Lead Developer

- Built a facial recognition and emotion detection API using **OpenCV** and **TensorFlow**.
- Designed a responsive dashboard to visualize real-time inference data from multiple camera streams.

SynthText NLP | Project Lead

- Fine-tuned a Transformer-based model for specialized legal document summarization.
- Integrated the model into a microservices architecture using **FastAPI** and **Docker**.

Education

M.S. in Computer Science (Artificial Intelligence) *Stanford University*

B.S. in Software Engineering *Massachusetts Institute of Technology (MIT)*

Certifications

- **AWS Certified Machine Learning – Specialty**
- **DeepLearning.AI TensorFlow Developer Professional Certificate**